

Margarita Montañez Davenport

DECENTRALIZED SYSTEMS ARCHITECT
WEB3 ENGINEER · APPLIED RESEARCHER

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- › github.com/Future-Systems-Lab/Future-Systems-Lab-profile
- › americaoutloud.news/author/meg-montanez-davenport

BLOCKCHAIN & WEB3

SolidityHardhatRemix IDE

ERC-20/721Brave WalletEIP-6963

window.ethereumDIDs / SSIIPFS

FilecoinMerkle DAGsZK Proofs

libp2pBlockscout

ENGINEERING & DATA

Next.jsPostgreSQL 17PythonFlask

Streamlit systemd REST APIs Vercel

SSH Ed25519Git CI/CDAgile/Scrum

Tableau

HEALTHCARE & INFORMATICS

HL7FHIROMOPHIPAAEHR Mapping

Clinical Data ModelsOrthomolecular

ORGANIZATIONS

- › INCOSE — Systems Engineering
- › AMIA — Medical Informatics
- › HIMSS — Health IT
- › ISC² — Cybersecurity
- › PDMA — Product Management
- › ISOM — Orthomolecular Medicine
- › NANP — Nutrition Professionals

CERTIFICATIONS

- › Certified Blockchain Healthcare Professional (CBHP) — Blockchain Council · Apr 2025
- › Certified Smart Contract AuditorBlockchain Council · Jan 2026
- › Health Informatics — Johns HopkinsMay 2025
- › Web3 & Blockchain — INSEADApr 2025

CISA — PacktAug 2025

SSI Master — DizmeIDMar 2025

Medical Neuroscience — Duke SOMApr 2025

Health Informatics on FHIR — Georgia TechMar 2025

Business Analyst — IBMJun 2025

Data Analytics · UX Design · PM — GoogleJun 2025

IPFS · Filecoin · libp2p · Merkle DAGsIPFS / Filecoin Foundation / ProtoSchool · 2025

NC Accident & Health Insurance ProducerPre-licensing completed Apr 2026, exam scheduled

National Provider Identifier (NPI): 1497696264Naturopath (175F0000X), NPPES, 2026

SUMMARY

I design and deploy decentralized infrastructure for sovereign data governance — using regulated behavioral health as a deliberate proving ground for architectures that scale across complex domains. As a long-term healthcare practitioner with experience operating within HIPAA regulatory frameworks as a licensed clinical provider, I built within compliance frameworks before architecting beyond them — and FSL's infrastructure operates outside HIPAA scope by architectural design, holding no PHI. As founder of Future Systems Lab, I move from concept to production: on-chain deployments, EIP-191 wallet-gated consent architecture, and infrastructure designed to operate outside HIPAA scope by holding no PHI — a stronger architectural position than compliance, because the regulation doesn't apply when the system never holds the protected data. I think in systems, across disciplines, at the intersection of governance and user experience. Nine Ethereum smart contracts deployed and verified on Sepolia — every FSL system is in production. Pursuing a Doctor of Engineering to formalize applied research methodology and advance systems leadership at the intersection of blockchain, AI, and decentralized governance.

ENGINEERING EXPERIENCE

Lead Engineer — SovereignLedger Sovereign Health Governance Infrastructure

2026 — Present

Sepolia: [0x4afA577fA914068451e0Aa97b61F2396Of02aCc4](#)

- › Engineered a full PostgreSQL session governance schema spanning participants, sovereign guides, sessions, attestations, documents, audit logs, consent grants, and compliance log tables.
- › Built Phase 3 SMS-based approval-gated automation with case-insensitive approve/deny/defer, 4-hour timeout, and max-defer logic — deployed to production VPS via systemd.
- › Architected SovereignLedger as a standalone blockchain governance layer for session governance, attestation, and participant-controlled record verification — integrates with any payment pipeline (Crypto → XRP → Sovereign Navigation) and deployable independently of any single platform.

Practice Operations Lead — Riccobene Associates

2019 — 2025

70+ office practice network, North Carolina

- › Led operational scaling across 70+ office practice network, producing 64% per-provider production growth (\$380,710 → \$624,640 average per provider) through team coordination, workflow optimization, and cross-location process standardization.
- › Designed and deployed two-tier production-share compensation model adopted company-wide — 27% standard tier (network-wide rollout) and 33% architect tier (system designer compensation).
- › Per-provider share of total office production grew from 30.4% to 38.1% — system redesign enhanced overall office economics rather than redistributing existing revenue.
- › Recruited directly by CEO for C-suite engagement; maintained multi-year engagement while concurrently founding Future Systems Lab.

Lead Engineer — SovereignSession On-Chain Session Attestation

2026 — Present

Sepolia: [0xbEb13A36OC6FOC77Ea3af3650Ab9762a1B9965A1](#)

- › Designed and deployed wallet-native session attestation contract operating outside HIPAA scope by holding no Protected Health Information. EIP-191 wallet authentication for Sovereign Guide and Participant — no name fields, no PII captured by FSL infrastructure.
- › On-chain SessionStarted and SessionCompleted events with timestamps and duration only — immutable proof that a session occurred between two consenting wallets without revealing session content.
- › Phase 5 doctoral research (post-ASU acceptance): Two-Party Wallet-Signed Mutual Authentication with Client-Side Encrypted Session Recording. Advances SovereignSession from

single-party attestation (guide-initiated, Phase 1 deployed; Phases 2–4 scaffolded/specified) to bilateral cryptographic consent — both Sovereign Guide and Participant submit EIP-191 signatures before session start, verified on-chain. Client-side AES-256-GCM encrypted video, IPFS-pinned storage, on-chain hash proof, participant-controlled decryption keys. No competitor (Doxy.me, Daily.co, Zoom) implements two-party wallet-signed mutual consent with client-side encrypted recording and on-chain attestation.

Lead Engineer — AlchemistForge Permissionless On-Chain Shadow Integration Primitive

2026 – Present

Sepolia: [0xE092336F8f5082e57CcBb341A110C20ad186A324](#) · [alchemistforge.io](#)

- › Designed and deployed a fully permissionless smart contract for voluntary on-chain Jungian shadow integration — no admin, no owner, no pause mechanism, no token economics. Architecture covered by U.S. Provisional Patent 64/063,037.
 - › Two-function design: `alchemize()` records voluntary shadow disclosure; `celebrate()` marks integration completion. Both emit publicly queryable events. Zero PII stored on-chain — only pseudonymous wallet address, voluntary self-authored text, and timestamps.
 - › Live deployment integrates Brave Wallet + EIP-191 signature for participant authentication, public Blockscout verification, and "verify your contribution" UX inviting any visitor to inspect their own on-chain record.
 - › Novel celebration animation: Ophiuchus (suppressed 13th zodiac) traverses celestial firmament with trailing sparkle stream while the 12 standard zodiac glyphs burst from center — symbolic representation of integrated shadow material surfacing.
 - › Scaffolded mental-health-on-chain as a five-entry public smart contract awareness series; deployed 24/7 campaign orchestration across Dev.to, Twitter, Discord, GitHub, and Telegram. Recorded on-chain case study as Participant #1.
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Founder & Systems Architect — Future Systems Lab

2023 – Present

[github.com/Future-Systems-Lab/Future-Systems-Lab-profile](#)

- › Deployed and verified nine Ethereum smart contracts on Sepolia testnet: [HypnoNeuroToken \(HNT\)](#), [EHTv2](#), [MindMasteryNFT](#), [SovereignLedger](#), [AlchemistForge](#), [BenevolenceFund](#), [SovereignAchievement](#), [SovereignSession](#), and [NeuroBalanceConsent](#).
 - › Architected a 17-agent Master Command Center spanning Audit, Smart Contract, Backend Build, Frontend/UI, Compliance, Security, Testing/QA, and Research agents with full dropdown panel orchestration.
 - › Deployed production VPS automation stack (clawdbot) on OpenClaw VPS: Flask approval server, Telegram bot, Streamlit analytics dashboard, and two systemd services running continuously.
 - › Architected EIP-191 wallet-signed consent as unified authentication primitive across all 5 FSL platforms — single cryptographic signature serves as identity verification, informed consent, and session authorization. This integrated pattern is covered by U.S. Provisional Patent 64/063,037 and is the subject of a manuscript in preparation for Blockchain in Healthcare Today.
 - › Engineered EncryptAI — a 19-step canonical monorepo (frontend/, backend/, labs/, contracts/, game/) with Brave Wallet authentication via window.ethereum + EIP-191, PostgreSQL 17, and blockchain writes via viem — [Step 2 deployed as Secure Health Login](#).
 - › Established SSH-authenticated signed commits, GitHub org governance, and Rights Reserved licensing policy across all FSL repositories.
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Healthcare Data Architect — HypnoNeuro Mental Wellness Platform

2023 – Present

HypnoNeuro · EncryptHealth · SovereignLedger · NeuroBalance Watch — modular systems engineered to operate independently or as an integrated ecosystem

- › Architected and deployed a tier-based therapeutic game framework — currently operating with three flagship demos (one per level: L1 Theta Wave/GABA, L2 Biochemical/Serotonin, L3 Original Frequency/Endocannabinoid) demonstrating the full participant flow: wallet gating, HNT token rewards, on-chain achievement minting, and NFT-based tier access. Architecture supports expansion to full catalog and roadmap targets VR-mediated avatar circle integration.
- › Architected dual-chain healthcare session anchoring — every session cryptographically committed to both Ethereum Sepolia and XRPL testnet, generating a tamper-evident audit trail for sovereign health records.
- › Built ISOM-aligned 4-tier provider credentialing system (Platinum/Gold/Silver/Verified) with automated license verification via Verifiable.com and AI-assisted credential review using Claude Vision.
- › Implemented Walsh Protocol AI recommendation engine — biotype classification across 5 neurotransmitter systems (Trp, Tyr, Cortisol, Dopamine, Epi/NorEpi) with plant-based nutrient protocol generation.
- › Deployed avatar-based anonymous session rooms via Daily.co WebRTC — wallet-connected, provider-facilitated video sessions with sessionStorage-persisted character customization and SovereignLedger billing anchor.
- › Designed white-label operator licensing model with three commercial tiers (Starter \$497/mo, Growth \$1,497/mo, Enterprise custom) — full registration flow, Telegram-notified applications pipeline, and operator config abstraction layer.
- › Completed 35-step canonical build across 7 sprints — all five platforms live on Vercel and interconnected: HypnoNeuro, EncryptHealth, SovereignLedger, NeuroBalance, Provider Portal, and FSL Command Center.
- › Sprint 1–3: Core platform buildout — therapeutic game framework, Brave Wallet + XRPL dual-wallet auth, PostgreSQL schema, FHIR R4 export, XRPL session anchoring, NFT gating, HNT token rewards, provider portal backend.
- › Sprint 4–5: Avatar Session Rooms (Daily.co WebRTC), AlchemistForge HNT bridge, endocannabinoid baseline ML scorer, COA Lookup, Support Circles, Walsh Protocol recommendation engine, wellness scoring, XRPL wallet connect.

